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JAVA PROJECT REPORT ON

DIGITAL COMPLAINT REGISTRATION AND STATUS MONITORING SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled "**DIGITAL COMPLAINT REGISTRATION AND STATUS MONITORING SYSTEM"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **MUGILAN R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

(SHASTIKA L)

CERTIFICATE

This is to certify that the project entitled **"DIGITAL COMPLAINT REGISTRATION AND STATUS MONITORING SYSTEM"** is a **bona fide work** carried out by **SHASTIKA L B.Sc IT 'B' (2025-2028)**, **INFORMATION TECHNOLOGY**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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ABSTRACT

The Digital Complaint Registration and Status Monitoring System is a software application designed to streamline the process of registering, managing, and tracking complaints in an organization or institution. In many organizations, complaints are often handled manually, leading to delays, lack of transparency, and poor record management. This project aims to provide an efficient and automated solution to overcome these challenges.

The system allows users to register complaints by entering relevant details such as complaint type, description, and date. Once submitted, each complaint is assigned a unique ID, which can be used to track its status. The system enables administrators to view, update, and resolve complaints, while users can monitor the progress of their complaints in real time.

The application can be developed using programming technologies such as Core Java, along with concepts like classes, objects, methods, file handling or database integration, conditional statements, loops, and exception handling. It can be implemented as a console-based system or enhanced with a graphical user interface for better usability.

The primary objective of this system is to improve efficiency, ensure transparency, reduce response time, and maintain organized records of complaints. It provides a structured approach for handling issues and enhances communication between users and administrators.

This project demonstrates the practical implementation of software development concepts and serves as a foundation for building more advanced complaint management systems in the future.

1. INTRODUCTION

In today's digital era, effective complaint management systems are essential for organizations to maintain transparency, accountability, and customer satisfaction. Traditionally, many institutions handle complaints manually through written forms, phone calls, or in-person reporting. This manual approach is time-consuming, prone to errors, and often results in poor tracking and delayed responses to complaints.

With the advancement of information technology, digital complaint management systems have become crucial for improving efficiency and responsiveness. A computerized system enables users to register complaints online, track their status in real time, and receive timely updates, thereby enhancing user experience and organizational effectiveness.

The **Digital Complaint Registration and Status Monitoring System** is a Java-based application designed to automate the process of complaint handling. The system allows users to register complaints by entering relevant details, after which the system generates a unique complaint ID. Users can use this ID to track the progress and status of their complaints. The system also enables administrators to manage, update, and resolve complaints efficiently.

This project is developed using Core Java programming concepts such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a console-based program or by using Java Swing for a graphical user interface (GUI).

The primary goal of this system is to simplify complaint handling, ensure faster resolution, and maintain proper records of all complaints. This project also demonstrates how Java programming can be applied to develop practical real-world applications that improve service quality and operational efficiency.

1.1 Objective

The main objective of the **Digital Complaint Registration and Status Monitoring System** is to develop a simple and efficient application using Core Java that automates the process of complaint registration, tracking, and management.

The specific objectives of this project are:

- To design and develop a Java-based complaint registration system.
- To enable users to register complaints easily through a digital platform.
- To automate the tracking and updating of complaint status.
- To reduce manual errors and delays in complaint handling.
- To provide a user-friendly interface for both users and administrators.
- To demonstrate the implementation of Core Java concepts such as classes, objects, methods, loops, and conditional statements.
- To maintain a structured and organized record of complaints and their statuses.

1.2 Overview

The Digital Complaint Registration and Status Monitoring System is a menu-driven application developed using the Java programming language. The system provides a digital platform where users can register their complaints and track their status efficiently.

The system allows users to enter complaint details such as name, complaint type, description, and date. Once a complaint is submitted, the system generates a unique complaint ID for future reference. Users can use this ID to check the current status of their complaint at any time.

The application enables administrators to view all registered complaints, update their status (such as pending, in progress, or resolved), and manage complaint records effectively. This reduces manual work and ensures better organization of complaint data.

The system simplifies the complaint handling process by eliminating paperwork and improving response time. The application can be implemented as a console-based program or a graphical user interface using Java Swing.

This project demonstrates the practical use of Java programming in developing real-world applications for organizations, institutions, and service-based systems.

1.3 Features

The Digital Complaint Registration and Status Monitoring System includes the following main features:

- Complaint Registration – Allows users to register complaints by entering necessary details such as name, complaint type, and description.
- Unique Complaint ID Generation – Automatically generates a unique ID for each complaint for easy tracking.
- Status Tracking – Enables users to check the current status of their complaints using the complaint ID.
- Complaint Management – Allows administrators to view, update, and manage complaints efficiently.
- Status Update – Admin can update complaint status (e.g., pending, in progress, resolved).
- Data Storage – Maintains a structured record of all complaints for future reference.
- User-Friendly Interface – Provides a simple and easy-to-use menu-driven interface.
- Quick Processing – Reduces the time required for complaint registration and tracking.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the system requirements are clearly identified. This stage helps in understanding how the system should function and what features are necessary to solve the problem effectively.

The analysis phase focuses on identifying the limitations of the existing complaint handling system and proposing a computerized solution that improves efficiency, accuracy, and transparency.

2.1 Problem Definition

In many organizations and institutions, complaints are often handled manually through paper forms, phone calls, or in-person reporting. This traditional approach creates several challenges.

Some common problems include:

- Errors in recording complaint details
- Time-consuming complaint registration process
- Difficulty in tracking complaint status
- Lack of proper record management
- Delayed response and resolution
- Poor communication between users and administrators

Manual complaint handling systems increase the workload for staff and often lead to dissatisfaction among users due to delays and lack of transparency.

Therefore, there is a need for an automated system that allows users to register complaints easily, track their status, and ensures faster and more efficient resolution.

The **Digital Complaint Registration and Status Monitoring System** is designed to solve these problems by providing a computerized solution that streamlines the complaint management process.

2.2 Project Overview

The **Digital Complaint Registration and Status Monitoring System** is a Java-based application developed to simplify the process of complaint registration and tracking.The system provides a menu-driven interface where users can enter complaint details such as name, type of complaint, and description. After submission, the system generates a unique complaint ID that can be used to track the complaint status.

The system includes the following functionalities:

- Registering complaints with necessary details
- Generating a unique complaint ID
- Viewing complaint status using the complaint ID
- Managing and updating complaints by the administrator
- Maintaining records of all complaints

This project is implemented using Core Java programming concepts and can be developed as either a console-based program or a Java Swing GUI application.

2.3 Scope and Limitations

Scope

The scope of the Digital Complaint Registration and Status Monitoring System includes the following:

- Registering complaints through a digital platform
- Generating a unique complaint ID for tracking
- Allowing users to check complaint status
- Managing and updating complaints by administrators
- Maintaining organized records of complaints
- Demonstrating the practical use of Java programming concepts

The system can be used in:

- Educational institutions (schools and colleges)
- Government offices
- Corporate organizations
- Customer service centers
- Public service departments

Limitations

The system has certain limitations, including:

- The system is designed for basic complaint management and may not support complex workflows
- It may not include advanced security features such as encryption or multi-level authentication
- The system depends on manual input by users and administrators, which may lead to incorrect data entry
- Limited scalability for very large organizations with high volumes of complaints
- If implemented as a console-based application, it may not provide an advanced graphical user interface

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Digital Complaint Registration and Status Monitoring System is designed as a menu-driven application where users can register complaints, track their status, and administrators can manage and update complaints efficiently.
The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the Digital Complaint Registration and Status Monitoring System are explained below:

1. Complaint Registration Module

This module is responsible for allowing users to register their complaints by entering necessary details.
Functions of this module include:

- Accepting user details (name, contact, etc.)
- Accepting complaint type and description
- Storing complaint information in the system

Example Input:
Name: John
Complaint Type: Network Issue
Description: Internet not working properly
Once the complaint is submitted, it is stored for further processing.

2. Complaint ID Generation Module

This module generates a unique complaint ID for each registered complaint.
Functions of this module include:

- Automatically generating a unique ID
- Assigning the ID to each complaint
- Displaying the ID to the user for future reference

Example:
Complaint Registered Successfully!
Your Complaint ID: C1023
This ID helps users track their complaint status easily.

3. Status Tracking Module

This module allows users to check the current status of their complaints.
Functions of this module include:

- Accepting complaint ID as input
- Retrieving complaint details from the system
- Displaying current status (Pending / In Progress / Resolved)

Example Output:
Complaint ID: C1023
Status: In Progress

4. Complaint Management Module

This module is mainly used by the administrator to manage complaints.
Functions of this module include:

- Viewing all registered complaints
- Updating complaint status
- Managing complaint records

The administrator can change the status based on progress.

5. Report Generation Module

This module helps in displaying or maintaining records of complaints.
The report includes:

- Complaint ID
- User details
- Complaint type
- Status

This helps in keeping track of all complaints systematically.

6. Exit Module

This module allows the user to exit the application safely after completing all operations.
When the user selects the exit option, the system terminates the program.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Digital Complaint Registration and Status Monitoring System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1 Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of complaint handling processes.

The main problems identified in manual complaint systems are:

- Manual recording may lead to errors in complaint details
- Registering complaints manually takes more time
- Difficulty in tracking complaint status
- Lack of proper record maintenance
- Delayed response and resolution

Based on these problems, an automated system was designed

4.2 System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Complaint Registration Module
- Complaint ID Generation Module
- Status Tracking Module
- Complaint Management Module
- Report Generation Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3 Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

- Creation of classes and methods
- Implementation of menu-driven logic
- Writing code for complaint registration and tracking
- Handling user input

- Displaying complaint status and reports

Core Java concepts such as classes, objects, loops, conditional statements, and methods were used during development.

4.4 Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Register complaint	Complaint stored successfully
Generate complaint ID	Unique ID generated
Check status	Correct status displayed
Update status	Status updated properly
Invalid input	Error handled correctly

This phase ensures that the program runs without errors and produces accurate results.

4.5 Result and Documentation

After successful testing, the results of the system are documented. The system allows users to register complaints and track their status efficiently.The results are demonstrated through program execution and screenshots showing the input and output of the system.

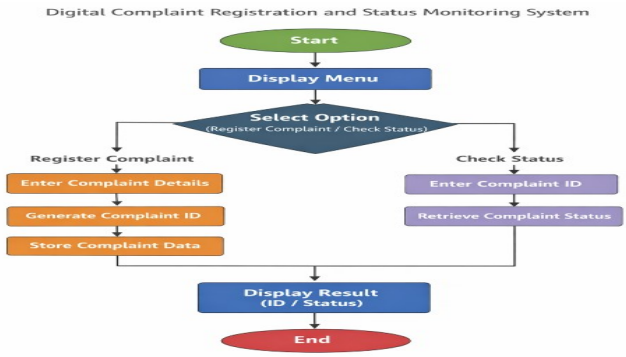
Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6 Algorithm

Algorithm for the **Digital Complaint Registration and Status Monitoring System**:

1. Start the program.
2. Display the main menu (Register Complaint / Check Status / Exit).
3. If the user selects "Register Complaint":
    - Enter user details and complaint description
    - Generate a unique complaint ID
    - Store complaint details
4. If the user selects "Check Status":
    - Enter complaint ID
    - Retrieve complaint details
    - Display complaint status
5. If the admin updates status:
    - Select complaint
    - Update status (Pending / In Progress / Resolved)
6. Repeat the process if required.
7. End the program.

4.7. Flowchart



5. SOURCE CODE

```
import java.util.*;

class ComplaintSystem {

    static Scanner sc = new Scanner(System.in);
    static Map<String, String> complaints = new HashMap<>();
    static int idCounter = 1001;

    public static void main(String[] args) {

        int choice;

        do {
            System.out.println("\n--- DIGITAL COMPLAINT SYSTEM ---");
            System.out.println("1. Register Complaint");
            System.out.println("2. Check Status");
            System.out.println("3. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            sc.nextLine(); // consume newline

            switch (choice) {
```

```
            case 1:
                registerComplaint();
                break;
            case 2:
                checkStatus();
                break;
            case 3:
                System.out.println("Exiting system...");
                break;
            default:
                System.out.println("Invalid choice!");
        }

    } while (choice != 3);
}

// Complaint Registration
public static void registerComplaint() {
    System.out.print("Enter your name: ");
    String name = sc.nextLine();

    System.out.print("Enter complaint: ");
    String complaint = sc.nextLine();

    String complaintId = "C" + idCounter++;
    complaints.put(complaintId, "Pending");

    System.out.println("Complaint Registered Successfully!");
    System.out.println("Your Complaint ID: " + complaintId);
```

```
    }

    // Status Checking
    public static void checkStatus() {
        System.out.print("Enter Complaint ID: ");
        String id = sc.nextLine();

        if (complaints.containsKey(id)) {
            String status = complaints.get(id);
            System.out.println("Complaint Status: " + status);
        } else {
            System.out.println("Invalid Complaint ID!");
        }
    }
}
```

1. INPUT AND OUTPUT SCREENS

COMPILATION SCREEN

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26100.4351]
(c) Microsoft Corporation. All rights reserved.

D:\java-class>javac  ComplaintSystem.java

D:\java-class>java  ComplaintSystem
```

INPUT SCREEN

```
C:\Windows\System32\cmd.e

D:\java-class>javac ComplaintSystem.java

D:\java-class>java ComplaintSystem

--- DIGITAL COMPLAINT SYSTEM ---
1. Register Complaint
2. Check Status
3. Exit
Enter your choice: 1
Enter your name: shasti
Enter complaint: shas
Complaint Registered Successfully!
Your Complaint ID: C1001
```

OUTPUT SCREEN

```
--- DIGITAL COMPLAINT SYSTEM ---
1. Register Complaint
2. Check Status
3. Exit
Enter your choice: 1
Enter your name: shasti
Enter complaint: shas
Complaint Registered Successfully!
Your Complaint ID: C1001

--- DIGITAL COMPLAINT SYSTEM ---
1. Register Complaint
2. Check Status
3. Exit
Enter your choice: 2
Enter Complaint ID: C1001
Complaint Status: Pending

--- DIGITAL COMPLAINT SYSTEM ---
1. Register Complaint
2. Check Status
3. Exit
Enter your choice: 3
Exiting system...
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Digital Complaint Registration and Status Monitoring System was developed to automate the process of complaint handling in organizations. The system allows users to register complaints, generate a unique complaint ID, and track the status of their complaints efficiently.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully manages complaint registration and enables users to check the status of their complaints.

The system was tested with different inputs to ensure that complaint details are stored correctly and status tracking works as expected. The results show that the system reduces manual effort, improves record management, and enhances transparency in complaint handling.

7.2 Conclusion

The Digital Complaint Registration and Status Monitoring System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world application.

The system automates complaint registration and tracking, reduces manual errors, and provides a faster and more efficient way to manage complaints. It simplifies the complaint handling process by maintaining organized records and allowing users to monitor complaint status easily.

This project helps students understand how Java programming can be used to develop useful software systems for real-world problems. The project can be further enhanced by adding features such as database connectivity, graphical user interfaces, user authentication, and online complaint submission portals.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

1. • https://www.oracle.com/java
2. • https://www.geeksforgeeks.org
3. • https://www.w3schools.com
4. • https://www.javatpoint.com